

Constrained Optimization I

Exercise 1

Two voltage sources connected in series with voltages x_1 [v] and x_2 [v] respectively are required to provide a constant voltage $V_L = 1$ [v] for a load. The cost for using the first one is x_1^2 . The voltage of the first one is upper bounded by 0.5 [v]. The cost of using the second one is $2x_2^2$. The voltage of the second one is not upper bounded.

(a) Formulate the optimization problem. Is it convex?

■ *Hint:* Remember that the voltages are always positive.

(b) Is the point $x = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$ [v] regular for the active constraints?

(c) Re-write the optimization problem such that all the constraints are equality constraints.

(d) Re-write the optimization problem such that all the constraints are inequality constraints.